

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

COG: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL 4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct:

(142511415)

I Karish Kumar S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Karish Kumar

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HP

C03- AT1

CASE STUDY

① CYBERSECURITY INTRUSION DETECTION SYSTEM:

a) Propositional Logic Representation,

F = Failed Login Attempts

U = Unusual IP Address

O = odd Login Time

H = High Data Transfer

B = Possible Brute Force Attack

C = Possible Account Compromise

D = Possible Data Exfiltration

Rules:

$$1. F \wedge U \rightarrow B$$

$$2. F \wedge O \rightarrow C$$

$$3. U \wedge H \rightarrow D$$

Facts for Session X:

* F = True

* U = True

* H = True

b) Forward chaining.

Initial facts:

F, U, H

Step 1 : Apply Rule 1

$$F \wedge U \rightarrow B$$

Since F = True & U = True:

B = True

→ possible Brute Force Attack

Step 2 : Apply Rule 2

$$F \wedge O \rightarrow C$$

D is not given

Therefore, Rule 2 is cannot fire.

Step 3: Apply Rule 3

$$U \wedge H \rightarrow D$$

Since $U = \bar{T}ue$ & $H = \bar{T}ue$

$$D = \bar{T}ue$$

→ Possible Data Exfiltration
Final Derived alerts:

- * Possible Brute force attack.
- * Possible Data Exfiltration.

② Backward chaining.

Goal D = Possible Data Exfiltration

Rule that concludes D:
 $U \wedge H \rightarrow D$

Goal: D

Rule 3: $U \wedge H \rightarrow D$

U
Fact True
H
Fact True

Both subgoals are facts in the KB:
* $U = \bar{T}ue$ ✓

* $H = \bar{T}ue$

Therefore,

$$D = \bar{T}ue$$

Hence, X session indicates Data Exfiltration.

REPORT PRIORITY LANDING SYSTEM :

A) Unification.

Rule 1 :

$$\forall x [\text{Aircraft}(x) \wedge \text{Emergency Declared}(x) \rightarrow \text{Priority clearance}(x)]$$

Given facts

* Aircraft (Flight z)

* Emergency Declared (Flight z)

Match :

Aircraft(x) with Aircraft (Flight z)

Therefore ,

$$x = \text{Flight z}$$

Sub ,

$$\theta = \{ x / \text{Flight z} \}$$

Next,

Emergency Declared(x) becomes :

Emergency Declared (Flight z)

which is also a fact.

$$\theta = \{ x / \text{Flight z} \}$$

The rule concludes :

Priority clearance (Flight z)

B) Resolution Proof of Delay.

Rule 1 :

$$\text{Aircraft}(x) \wedge \text{Emergency Declared}(x) \rightarrow \text{Priority clearance}(x)$$

CNF :

$$\neg \text{Aircraft}(x) \vee \neg \text{Emergency Declared}(x) \vee \text{Priority clearance}(x)$$

Rule 2 :

$$\text{Priority clearance}(x) \rightarrow \text{Runway Reserved}(x)$$

CNF :

$\neg$ Priority Clearance (x) $\vee$ Runway Reserved (x)

Rule 3:

Aircraft (x) $\wedge$ Queued After (x, y) $\wedge$ Runway Reserved (y) $\rightarrow$ Delay (x)

CNF:

$\neg$ Aircraft (x) $\vee \neg$ Queued After (x, y) $\vee \neg$ Runway Reserved (y) $\vee$ Delay (x)

We derive:

Delay (Flight x)

Finally resolve with,

$\neg$ Delay (Flight x)

* Empty Clause

Hence, Delay (Flight x) is proven True.

2) Two simultaneous emergency aircraft.

The current KB can handle multiple craft individually because Rule 1 is Universally quantified.

Rule 1:

Priority Clearance (Flight A)

Rule 2:

Runway Reserved (Flight A)

$\therefore$ KB does not explicitly handle runway conflicts between two simultaneous emergency aircraft.

$\forall x \forall y [(Priority\ Clearance(x) \wedge Priority\ Clearance(y) \wedge x \neq y) \rightarrow Runway(x, y)]$

Additional facts such as,

* Runway availability

* aircraft priority

* arrival time

* Runway assignment

Would be required.

1) CNF :

1. P1 :

Machine Overheating $\rightarrow$ Maintenance Needed

$\neg$ Machine Overheating $\rightarrow$ Maintenance Needed

2. P2 :

Maintenance Needed $\wedge$ High Priority Line $\rightarrow$ Schedule Immediate Repair

$\neg$ Maintenance Needed $\vee \neg$ High Priority Line $\rightarrow$ Schedule Immediate Repair

3. P3 :

$\neg$ Schedule Immediate Repair $\wedge$ High Priority Line $\rightarrow$ Production Risk

Schedule Immediate Repair $\vee \neg$ High Priority Line $\rightarrow \neg$ Production Risk

4. Facts :

Machine Overheating

High Priority Line

5) Resolution Refutation,

Goal : Schedule Immediate Repair

Negated goal : $\neg$ Schedule Immediate Repair

1. Machine Overheating

* ($\neg$ Machine Overheating $\vee$ Maintenance Needed)
 $\rightarrow$ Maintenance Needed

2. Maintenance Needed

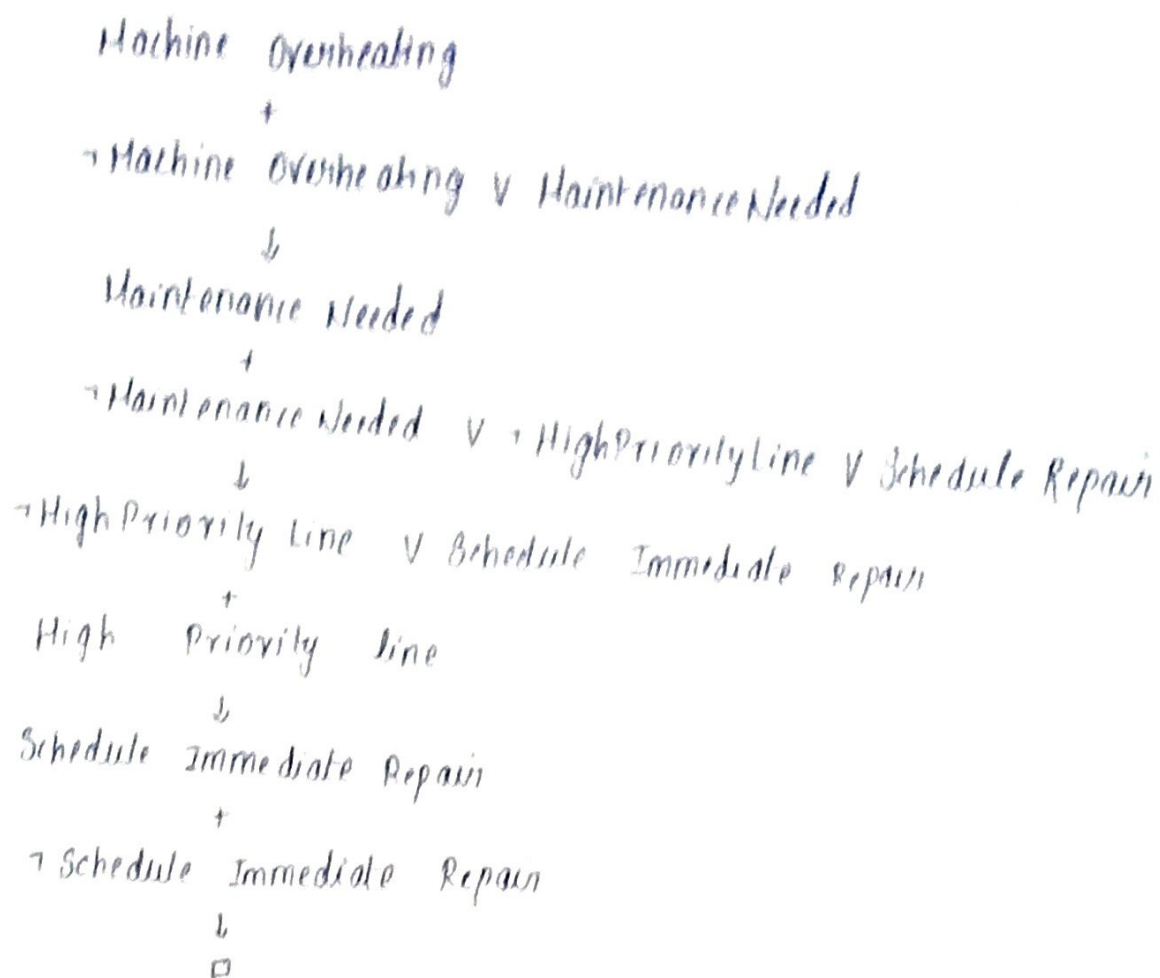
* ($\neg$ Maintenance Needed $\vee \neg$ High Priority Line $\vee$ Schedule Repair)
 $\rightarrow \neg$ High Priority Line $\vee$ Schedule Repair

3. Schedule Immediate Repair

* $\neg$ Schedule Immediate Repair

$\rightarrow$ Empty Clause

Resolution Tree:



② If Machine Overheating is Removed,

Without Machine Overheating, P1 cannot derive Maintenance Needed.
Therefore,

Maintenance Needed $\times \rightarrow$ Schedule Immediate Repair cannot be derived
Also, $\rightarrow$ Schedule Immediate Repair is not given as a fact.
Hence P3 cannot fire.

Production Risk cannot be logically concluded.

FINAL:

Removing Machine Overheating means neither schedule Immediate Repair nor production risk can be derived from the given KB.

MINEFIELD ROBOT KNOWLEDGE AGENT :

a) Belief state and Modus Ponens ,

Initial concepts

* Vibration [2,1]

* Smoke [1,1]

* Beacon [2,2]

Rules ,

1. Vibration [2,1] $\rightarrow$ Mine Adjacent [2,1]

2. Smoke [1,1] $\rightarrow$ Fire Adjacent [1,1]

3. Beacon [x,y] $\rightarrow$ Safe zone Marker [x,y]

4. $\neg$ Mine [x,y] $\wedge$ $\neg$ Fire [x,y] $\rightarrow$ Safe [x,y]

Modus Ponens ,

Step 1 :

Vibration [2,1]

Rule 1 :

Vibration [2,1] $\rightarrow$ Mine Adjacent [2,1]

Mine Adjacent [2,1] .

Step 2 :

Smoke [1,1]

Rule 2 :

Smoke [1,1] $\rightarrow$ Fire Adjacent [1,1]

Fire Adjacent [1,1]

Step 3 :

Beacon [2,2]

Rule 3 :

Beacon [2,2] $\rightarrow$ Safe Zone Marker [2,2]

Safe Zone Marker [2,2] .

b) Forward chaining,

Step 1:

Vibration $[2,1] \rightarrow$ Mine Adjacent $[2,1]$

Conclusion $\rightarrow$ A mine is adjacent to cell $[2,1]$.

Step 2:

Smoke $[1,1] \rightarrow$ Fire Adjacent $[1,1]$

Conclusion $\rightarrow$ A fire is adjacent to cell $[1,1]$

Step 3:

Beacon $[2,2] \rightarrow$ Safe Zone Marker $[2,2]$

Conclusion $\rightarrow$ $[2,2]$ is marked as the Safe Zone.

FINAL: The exact locations of the mine and fire cannot be determined from the given rules.

c) Path Safety.

Path: $[1,1] \rightarrow [2,1] \rightarrow [2,2]$

* At $[1,1]$, Smoke indicates a nearby fire.

* At $[2,1]$, Vibration indicates a nearby mine.

* At $[2,2]$, the Safe Zone Marker is present.

The KB does not prove that $[1,1]$ and $[2,1]$ are completely safe. Therefore, the robot cannot logically confirm this path as safe.

CONCLUSION: The path's not proven safe based on the available knowledge.

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding, misses key elements	Misinterprets or unable to understand problem	15
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	23
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors, incorrect final answer	No valid calculations	18
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	13

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